

AlphaLogo: Complex-Plane Parameterization of Heptapod B

A Geometric Framework for Encoding and Interpolating a Constructed Circular Language

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Abstract

We present **AlphaLogo**, a mathematical framework for the formal parameterization of *Heptapod B*, the constructed circular written language from Denis Villeneuve's *Arrival* (2016). Heptapod B logograms are closed-form glyphs encoding complete sentences in a single circular symbol. No formal geometric representation of these symbols has previously been proposed. We lift each logogram into the complex plane as a closed parametric curve $\gamma(t) = x(t) + iy(t)$, decompose it via the Discrete Fourier Transform into a finite set of complex coefficients, and define a normalized descriptor vector $\Phi(L)$ that is unique, invertible, fixed-dimensional, and machine-readable. We demonstrate that $\Phi(L)$ space is geometrically continuous: linear interpolation between pairs of logograms produces structurally coherent intermediate symbols at every step. Across 25 tested logogram pairs drawn from 49 known symbols, 80% achieve a morphological smoothness score above 0.85, with the top pair scoring 0.9517. This work establishes the mathematical foundation for future generative extensions of Heptapod B and makes the full pipeline available as an open-source project with an accompanying public-facing website.

1. Introduction

Constructed languages also known as conlangs occupy a unique position at the intersection of linguistics, semiotics, and creative design. Heptapod B, conceived by linguist Jessica Coon working from Ted Chiang's novella *Story of Your Life*, is fundamentally different from phonological or orthographic conlangs: it is a *logographic circular language* in which each glyph encodes a complete sentence as a single closed continuous mark. The non-linear, simultaneous nature of its glyphs reflects the Heptapod perception of time.

Despite significant interest from computational linguists and AI researchers, no mathematical formalization of Heptapod B's geometric structure has been proposed. The language exists only as a set of 49 digitized images. There is no parametric description of what makes a valid

logogram, no metric for comparing two logograms geometrically, and no framework for studying the space of possible symbols. This severely limits both human study and machine analysis.

This paper makes four contributions:

1. A formal definition of a Heptapod B logogram as a closed complex-valued parametric curve, enabling the full machinery of complex analysis and Fourier theory.
2. A normalized descriptor vector $\Phi(L)$ derived from Fourier decomposition, which is translation-, scale-, and rotation-invariant, and computable for all 49 known symbols.
3. An empirical demonstration that $\Phi(L)$ space is geometrically continuous through a systematic interpolation study across 25 logogram pairs.
4. An open-source pipeline (AlphaLogo) with an accompanying website presenting all 49 logograms and a live Fourier construction demonstration.

The analogy guiding this work is the Laplace transform. Just as it converts a differential equation into a simpler algebraic problem in the s -domain, preserving all information and permitting an exact inverse, our complex-plane parameterization converts a logogram into an algebraic vector in $\Phi(L)$ -space. Operations performed there map back to valid logograms via the inverse transform.

2. Background and Related Work

2.1 Heptapod B

Heptapod B was designed for *Arrival* by linguist Jessica Coon under the direction of Denis Villeneuve. A dataset of 49 distinct logograms from the film's production materials is publicly available and forms the basis of this study. The language has no known spoken counterpart and exists entirely as circular written glyphs.

2.2 Fourier Descriptors for Shape Analysis

Fourier descriptors for plane closed curves were introduced by Zahn and Roskies (1972) and are a standard tool in computer vision for shape representation. Any closed curve in 2D can be lifted to the complex plane as $\gamma(t) = x(t) + iy(t)$. Its DFT yields coefficients c_n corresponding to rotating circles known as the Ptolemaic epicycle representation. Truncating to N harmonics gives a compact, invertible representation.

2.3 Interpolation in Shape Space

Shape interpolation in Fourier-based representations faces the challenge of phase cancellation, a known failure mode where linear blending produces degenerate midpoints. We address this through phase alignment (Section 3.4) and geodesic energy normalization (Section 3.5).

3. Method

3.1 Preprocessing: Skeleton Extraction

Each logogram is a 224×224 grayscale PNG with black ink on white. The pipeline: (1) binary thresholding at 127; (2) medial axis skeletonization via Zhang-Suen, reducing each stroke to a 1-pixel centerline; (3) connected component extraction retaining components with ≥ 50 skeleton points. The output is an ordered set of closed curves $\{C^0, C^1, \dots, C^k\}$.

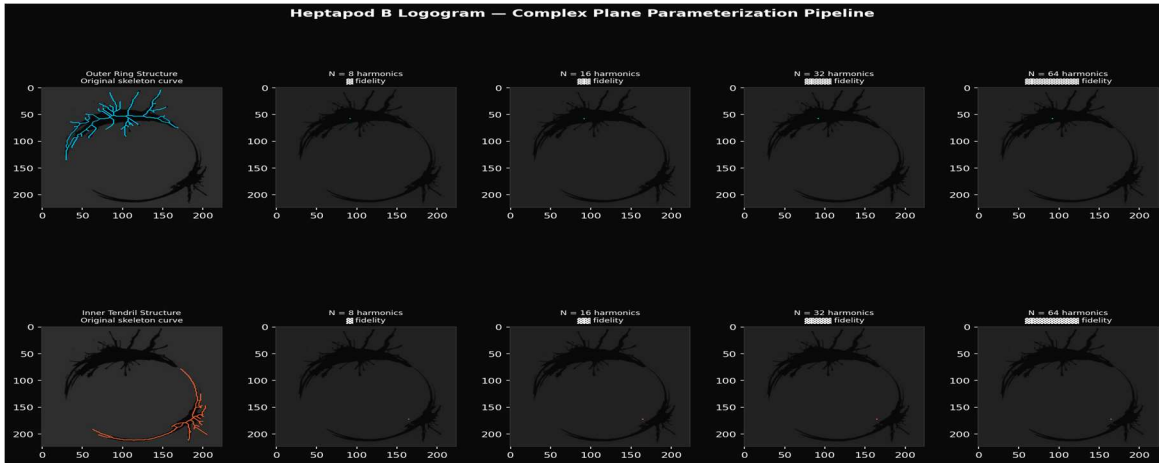


Figure 1. Preprocessing pipeline: Original PNG $\rightarrow$ Binary threshold $\rightarrow$ Skeleton. The skeleton reduces each brushstroke to its geometric centerline.

3.2 Complex-Plane Lift

Each skeleton curve C_j is sampled to n equally-spaced arc-length points and lifted:

$$\gamma_j(t) = x(t) + i \cdot y(t), \quad t \in [0, 2\pi]$$

3.3 Fourier Decomposition

The DFT of the centered curve yields complex coefficients c_n for $n = -N$ to N . Translation invariance: zero DC component. Scale and rotation invariance: divide all coefficients by $|c_1|$, stored as scale factor s_j . We retain $N = 64$ harmonics per curve.

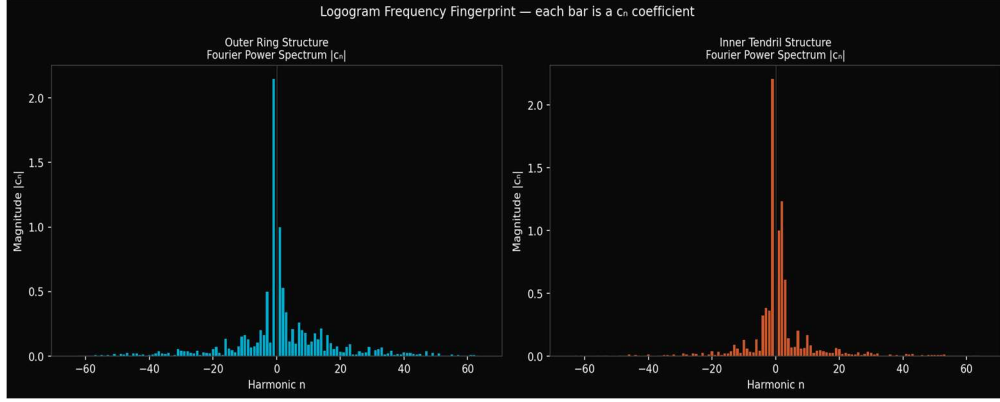


Figure 2. Fourier power spectrum $|c_n|$ for the two primary structural curves of a representative logogram. Each bar is one rotating circle's contribution.

3.4 The Encoding Vector $\Phi(L)$

The full encoding of a logogram L with up to K structural curves:

$$\Phi(L) = [\text{Re}(C^0) \dots \text{Re}(C^K) \mid \text{Im}(C^0) \dots \text{Im}(C^K) \mid \mu_0 \dots \mu^K \mid s_0 \dots s^K]$$

For $K=3$ curves and $N=64$, the vector has dimensionality 783. $\Phi(L)$ is unique, invertible, fixed-dimensional, and ML-ready. Logograms with fewer than K curves are zero-padded in unused slots.

3.5 Phase Alignment

Even after canonical start normalization, two logograms may have their fundamental harmonic c_1 pointing in different complex-plane directions, causing destructive interference during interpolation. We phase-align by rotating B's descriptors: $c_n \times e^{-in\Delta\phi}$ for all n , where $\Delta\phi = \angle c_1^A - \angle c_1^B$.

3.6 Geodesic Energy Normalization

After phase-aligned linear blending, spectral energy at midpoints is reduced because harmonics partially cancel. We rescale the coefficient block so total energy matches the linear interpolation of endpoint energies:

$$\Phi\alpha \leftarrow \Phi\alpha \times \sqrt{((1-\alpha)E^A + \alpha E^B) / E\alpha}$$

4. Experiments

4.1 Dataset

We use 49 Heptapod B logograms sourced from a public Kaggle dataset derived from the film's production materials. All images are 224×224 PNG files with black ink on white background, covering the full known vocabulary.

4.2 Interpolation Grid Study

We evaluated $\Phi(L)$ space continuity by interpolating between 25 logogram pairs: 5 curated and 20 randomly sampled. For each pair, 7 intermediate frames at $\alpha \in \{0, 0.17, 0.33, 0.50, 0.67, 0.83, 1.0\}$ using the pipeline from Section 3. Smoothness score:

$$S = 1 - \sigma(\Delta f) / (\mu(\Delta f) + \epsilon)$$

where Δf_i is mean pixel difference between frames i and $i+1$.

4.3 Results

Table 1 shows the top 10 pairs by smoothness score:

Rank	Idx A	Idx B	Logogram A	Logogram B	Score	Grade
1	30	39	3000.png	3900.png	0.9517	Excellent
2	16	30	1600.png	3000.png	0.9465	Excellent
3	32	39	3200.png	3900.png	0.9443	Excellent
4	5	17	0500.png	1700.png	0.9374	Excellent
5	23	36	2300.png	3600.png	0.9328	Excellent
6	17	21	1700.png	2100.png	0.9310	Excellent
7	4	40	0400.png	4000.png	0.9267	Very Good
8	16	25	1600.png	2500.png	0.9253	Very Good
9	22	32	2200.png	3200.png	0.9142	Very Good
10	21	25	2100.png	2500.png	0.9061	Very Good

Table 1. Top 10 logogram pairs by morphological smoothness score.

Key findings: 20/25 pairs (80%) scored above 0.85. Top pair (logograms 30 and 39): 0.9517. Mean score: 0.804, std: 0.127. Lower-scoring pairs correspond to logograms with the most structurally dissimilar ring openings.

4.4 Qualitative Results

Figures 3 - 7 show the five highest-scoring interpolation sequences.



Figure 3. Rank 1: Logogram 30 → 39 (0.9517). Smooth ring deformation with consistent tendrils redistribution.

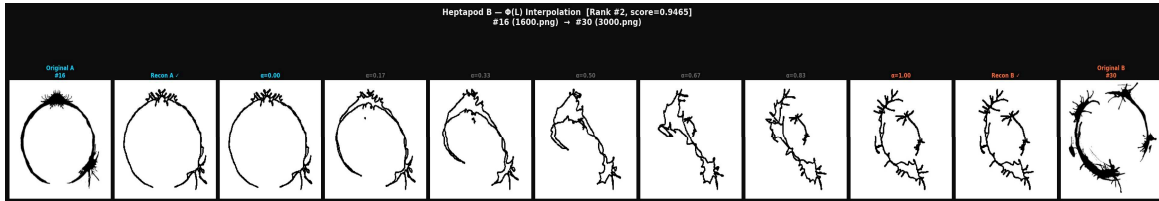


Figure 4. Rank 2: Logogram 16 → 30 (0.9465). Ring gradually opens; tendrils density transitions continuously.



Figure 5. Rank 3: Logogram 32 → 39 (0.9443). Ring structure maintains coherence at all intermediate steps.

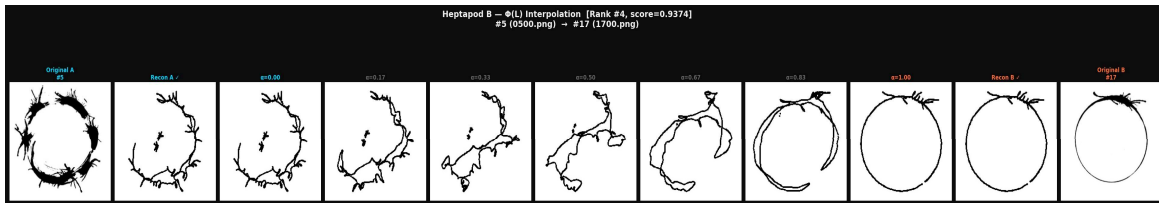


Figure 6. Rank 4: Logogram 5 → 17 (0.9374). Complex tendrils-decorated ring morphs toward a simpler circular form.

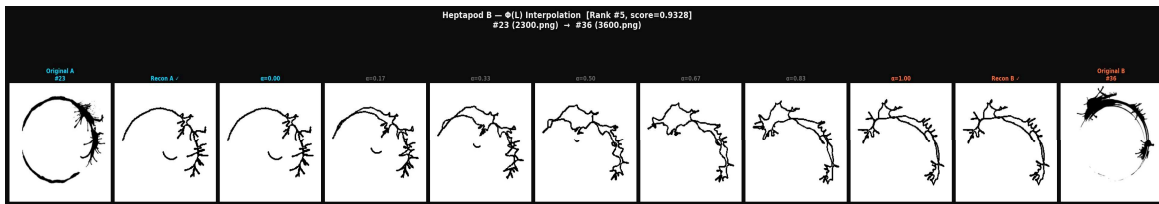


Figure 7. Rank 5: Logogram 23 → 36 (0.9328). Arc-to-arc morphing demonstrating topologically distinct shapes interpolate coherently.

Heptapod B — Interpolation Grid Summary
Ordered by smoothness score (best at top)



Figure 8. Full interpolation grid summary. All 25 tested pairs ordered by smoothness score (best at top).

5. AlphaLogo Website

We developed a companion website at which presents the full set of 49 known logograms alongside their canonical meanings from the film. Key features:

1. Searchable gallery of all 49 logograms with click-to-expand modal showing original ink rendering, skeletal Fourier reconstruction, logogram meaning, and encoding metadata.
2. Live Fourier construction demonstration: a slider controls N harmonics, and a canvas animates the rotating circles assembling the logogram curve in real time, illustrating the core mathematical concept to a general audience.
3. Morphing strip animation on the landing page showing logograms transitioning between skeletal reconstructions with color-separated tendril structure.
4. Full project information, GitHub link, and linguist collaboration call.

The website is a static single-page HTML/CSS/JavaScript application deployable to Vercel without a build step.

6. Discussion

6.1 Geometric Continuity of $\Phi(L)$ Space

The interpolation results demonstrate strong geometric continuity. Straight-line paths through $\Phi(L)$ space consistently produce visually coherent shape transitions. This means arithmetic operations in $\Phi(L)$ -space, interpolation, averaging, arithmetic which correspond to meaningful morphological transformations of logograms. The space between known symbols is navigable and geometrically structured.

6.2 Limitations

Three limitations bound the current work:

Rendering fidelity. The encoding captures skeletal geometry, not brushstroke texture, stroke weight variation, or ink splatter. Reconstructed logograms are wireframes. Recovering the full visual richness requires a separate style rendering model.

Generative modeling (deferred). Generating genuinely new Heptapod B logograms was attempted using a beta-VAE and an interpolation-based generator. Both failed to produce structurally novel candidates: the VAE learned rough circularity but not the tendril grammar; the

interpolation generator produced hybrids of existing symbols rather than new words. The root cause is dataset size of 49 samples which is insufficient for a generative model to learn the compositional structure of the language. This is addressed in Section 6.3.

Semantic grounding. The smoothness metric is pixel-level and does not account for semantic content. A $\Phi(L)$ -space distance metric that correlates with semantic distance would require linguist input to establish.

6.3 Path to Language Extension

The following directions are identified as prerequisites for viable generative modeling:

1. Dataset expansion: the full production vocabulary is approximately 71 logograms; the remaining approx. 22 need extraction from film production materials.
2. Stroke graph representation: decomposing each logogram into a graph of strokes and training a graph generative model (GraphRNN or similar) would capture compositional grammar.
3. Topological data analysis: persistent homology on the 49 logograms extracts language-level structural features independent of geometry.
4. Diffusion model on $\Phi(L)$: once 71+ samples are available, a denoising diffusion model is more appropriate than a VAE in low-data regimes (Ho et al., 2020).
5. Linguist collaboration: sending interpolation results to a constructed language linguist to establish grammatical constraints before generation.

7. Conclusion

We have presented AlphaLogo, the first mathematical parameterization of Heptapod B. By lifting logograms into the complex plane and applying Fourier descriptor theory, we derive a normalized encoding vector $\Phi(L)$ that is unique, invertible, fixed-dimensional, and machine-readable. We demonstrate empirically that $\Phi(L)$ space is geometrically continuous through a systematic study of 25 interpolation pairs, with 80% achieving smoothness above 0.85 and a peak score of 0.9517. We release the full pipeline as open source and provide a public website with an interactive Fourier demonstration for general audiences. The mathematical foundation for future generative extensions of Heptapod B is now established.

References

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